COMP0197 Applied Machine Learning Systems

Module handbook 2026/27 — Term 1 — 15 credits

1. Module overview

COMP0197 is about what happens to a machine learning model after the notebook is closed. We treat a trained model as one component inside a larger system and spend the term on the parts of that system which decide whether it survives contact with real users: data collection, evaluation design, deployment, monitoring, and the documentation that lets somebody else take the work over. The module is deliberately practical. You will spend more time reading other people's pipelines than deriving results on paper.

The module is convened by Dr A. Okonkwo and taught jointly with Dr L. Marchetti. Office hours are listed on the Moodle page and are held in person in the Malet Place Engineering Building.

2. Learning outcomes

By the end of the module you should be able to:

- Design an evaluation that reflects how a model will actually be used, including the failure modes that matter to the people affected by it.
- Build and instrument a deployment pipeline, and explain the trade-offs you made in doing so.
- Document a model so that a reader outside your team can judge whether it is fit for their purpose.
- Read and critique a published machine learning system, not just a published result.

3. Teaching pattern

Two hours of lectures and one two-hour laboratory session per week across Term 1. Laboratory sessions begin in the second week of term; the first week is lectures only. There is no teaching in the reading week commencing Monday 9 November 2026, although the laboratory machines remain available and project teams are expected to use that week for their own planning.

Attendance at laboratory sessions is recorded but is not itself assessed. Students who miss more than three sessions will be contacted by the departmental tutor.

4. Reading and resources

There is no single textbook. A reading list is published on Moodle and updated through the term. Two or three papers are set for each lecture and you are expected to have read at least one of them before the session. Laboratory work uses Python 3.11 on the department's teaching cluster; accounts are created automatically once your module registration is confirmed.

5. Assessment

The module is assessed entirely by coursework. There is no written examination. Four components contribute to the final mark and all four must be attempted for a mark to be returned. They are described below in the order they fall due.

5.1 Problem Set 1 (15%)

Six short exercises on evaluation design and metric selection, released in the fourth week of term. The problems are individual work and are marked out of sixty. There is no word limit; answers are expected to be concise, and most students write between a paragraph and a page per exercise. Submit by 4 December 2026, 16:00 through the Moodle submission point. Model solutions are released immediately after the deadline, so late submissions cannot be accepted under any circumstances.

5.2 Coursework 1: model card report (35%)

You will be given a trained model, its training data, and nothing else, and asked to produce a model card of no more than 2,500 words covering intended use, evaluation, limitations, and the questions you could not answer from the material provided. The last of those sections usually carries the strongest marks. This component is due in Week 5 and is submitted as a single PDF.

5.3 Group project: deployment case study (20%)

Working in teams of four, you will take one of the systems discussed in lectures and write a case study of how it is deployed, instrumented and monitored. The report should be around 4,000 words for the team as a whole, not per member. Teams are formed in the laboratory session of the third week and cannot be changed afterwards without the convenor's agreement. The report is due in UCL timetable week 14. Every member of a team receives the same mark unless the team asks for a contribution split, in which case the convenor will interview the team before marks are issued.

5.4 Showcase presentation (30%)

Each team presents its case study to the cohort and to two members of staff. Presentations are twelve minutes with eight minutes of questions, and the questions are marked. The showcase runs all day on Thursday 10 December 2026 in Roberts 110. The running order is drawn at random and published the preceding Friday, so be prepared to present at any point during the day.

6. Submission and late penalties

Unless stated otherwise above, coursework is submitted through Moodle and the departmental deadline convention of 16:00 applies. Work submitted up to two working days late is capped at the pass mark; work submitted after that receives zero. If something has gone wrong, use the extenuating circumstances process rather than emailing the convenor, and do so before the deadline wherever possible.

7. Feedback

Written feedback on Coursework 1 is returned within one calendar month of the deadline. Problem Set 1 is marked against a published scheme and individual feedback is brief by design. Presentation feedback is given verbally on the day and confirmed in writing afterwards.

8. Academic integrity

You may use generative AI tools for any component of this module provided you say where and how, in an appendix that does not count towards the word limit. Undeclared use is treated as a breach of academic integrity. Collaboration on the problem set is not permitted; collaboration within your team on the group project obviously is.